

Supplementary information (ESM_1)

Reciprocal-free divisors of prescribed order over finite fields and the saturation cost of cyclic quantum synchronizable codes

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This file contains the proof of Lemma 2.3, the Class 2 observations of Sect. 11.4 and the table of computational checks of Sect. 12.3 of the main article. Section, lemma and theorem numbers refer to the main article. “Li-Zhu” denotes Z. Li and S. Zhu, Int. J. Theor. Phys. 61, 251 (2022). The Python code is provided as ESM_2.zip.

S1. Proof of Lemma 2.3

Lemma 2.3 (dual containment). Let $\gcd(N, q) = 1$, and let C be a cyclic code of length N over F_q with generator polynomial g_C . Then

$$C^\perp \subseteq C \iff g_C(x) g_C^*(x) \mid x^N - 1.$$

Proof. Write $R_N = F_q[x]/(x^N - 1)$ and $x^N - 1 = g_C h_C$.

Step 1 (orthogonality as a polynomial condition). For $u, v \in R_N$ put $\bar{v}(x) = \sum_j v_j x^{-j} \in R_N$. The coefficient of x^s in $u(x) \bar{v}(x)$ is $\sum_j u_{j+s} v_j$, with indices modulo N . This is the inner product of v with the s -th cyclic shift of u . Since C is spanned by the cyclic shifts of g_C ,

$$v \in C^\perp \iff g_C(x) \bar{v}(x) = 0 \text{ in } R_N.$$

Step 2 (identification of $C^\perp$). Now $g_C \bar{v} = 0$ in R_N if and only if h_C divides $\bar{v}$ (as an element of R_N). The map $v \mapsto \bar{v}$ is the ring automorphism of R_N induced by $x \mapsto x^{-1}$. It carries $\langle h_C \rangle$ onto $\langle h_C(x^{-1}) \rangle = \langle h_C^* \rangle$, because $x^{\deg h_C}$ is a unit of R_N . Hence $C^\perp = \langle h_C^* \rangle$.

Step 3 (comparison of ideals). Since h_C^* is a monic divisor of $x^N - 1$, the inclusion $C^\perp \subseteq C$ holds if and only if $g_C \mid h_C^*$. Taking reciprocals, this is equivalent to $g_C^* \mid h_C$, and that is equivalent to $g_C g_C^* \mid g_C h_C = x^N - 1$. ■

S2. Class 2 computational observations (Sect. 11.4)

For the BCH class of Li-Zhu one has $n = (q^{2m} - 1)/(q - 1)$ and $N = 2n$. The observations below use an index rule reconstructed from

- the defining-set truncations printed in Example 3 of Li-Zhu;
- the ranges $1 \leq i \leq \delta - 2$ and $1 \leq j \leq \delta^* - 2$;
- the upper limits for δ and δ^* stated by Li-Zhu.

“Saturated” means $\text{ord}(h) = N$ under this reconstruction.

(q, m)	N	$\text{ord}_N(q) = w_0 = w$	tuples	saturated	$\text{ord}(h) < N$	range of s (saturated)
(3, 2)	80	4	75	75	0	8-24
(5, 2)	312	4	1,490	1,482	8	6-66
(7, 2)	800	4	10,269	10,263	6	8-124
(3, 3)	728	6	19,908	19,896	12	12-144

In every saturated tuple observed, $s > w$. These are computational observations under a reconstructed rule; they are not a theorem and they do not cover Class 2

exhaustively. The tuples with $\text{ord}(h) < N$ may be excluded by the exact hypotheses of Li-Zhu. The observed minimum $s = 6$ for $(q, m) = (5, 2)$ is below $2w = 8$, so the bound $s \geq 2w$ of Theorem 11.1 is not claimed for Class 2.

S3. Computational checks (Sect. 12.3)

The values w and w_0 were computed by a dynamic programme over the lcm-cover form (Proposition 4.2) and cross-checked against an independent brute force that enumerates sets of q -cyclotomic cosets directly, without using Proposition 4.2. The support-reduction enumerator for Theorem A allows distinct blocks to share a support, as in Sect. 6.6, where 45 and 75 both have support $\{3, 5\}$.

Check	Range (finite)	Checks	Failures
DP vs. independent coset brute force	15 values of q (even and odd), $N < 200$	1,999	0
Lemma 5.1	all prime powers $q < 128$; odd $m < 5000$	99,981	0
Theorem A	all prime powers $q < 128$; odd $N < 2000$	39,968	0
Formula (6.3) for w_0	as for Theorem A	39,968	0
Theorem B	all prime powers $q < 128$; odd two-prime $N < 20000$	195,405	0
Theorem C(i)	all prime powers $q < 128$; odd primes $p < r < 400$	30,079	0
Theorem C(ii)	explicit triples (p, r, q) with $p, r < 400$	594	0
Theorem 11.1(i) and the factor-degree structure	odd $q < 128$; primes $n < 400$	1,081	0
Total		409,075	0

The ranges include adversarial cases: repeated prime powers; three and four distinct primes; mixed 2-adic classes; blocks requiring helper primes; supports used by more than one block; and even q for Lemma 5.1 and Theorems A and B. The record is not an exhaustive verification over all inputs.

Instances from Li-Zhu. Generator degrees, $\text{ord}(h)$, dual containment and the exact $d(D)$ were computed directly for the three examples. In Example 3 the factor $m_0 = x - 1$ printed in g_1 and g_3 is incompatible with dual containment and with the printed dimensions; the computations omit it (Sect. 11.3).

S4. Code

The Python scripts are provided as ESM_2.zip; see its README.